Clover: Drone Diagnostics

Automated drone propulsion system health diagnostics

UNIVERSITY OF TORONTO

D.VISION AERIALS

Problem Context

Filming at a concert
Credit: D. Vision Aerials

- FPV drones are sometimes operated in densely populated areas across the GTA
- Even minor motor faults can cause crashes, risking injury and property damage
- Current checks rely on operators listening to motors for irregularities
- This process is subjective and difficult to verify
- Equipment failure could expose operators to up to $12.5M in liability

Opportunity

A portable system to verify the health and flight readiness of drone propulsion systems using objective pre-flight diagnostics.

Design Objectives

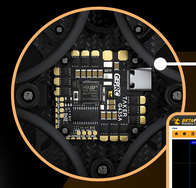
1. **Fault Diagnosis** — Detect, isolate, and classify propulsion system faults
2. **Diagnostic Reliability** — Maintain accurate, repeatable diagnostics under environmental noise
3. **Operational Integration** — Rapid diagnostics that integrate into standard pre-flight workflows
4. **Field Portability** — Portable, infrastructure-independent operation for outdoor field usage
5. **Diagnostic Traceability** — Record and retain time-stamped diagnostic results linked to the drone's components.

Drone

GEPRC Cinelog25 V2 HD O3 Quadcopter

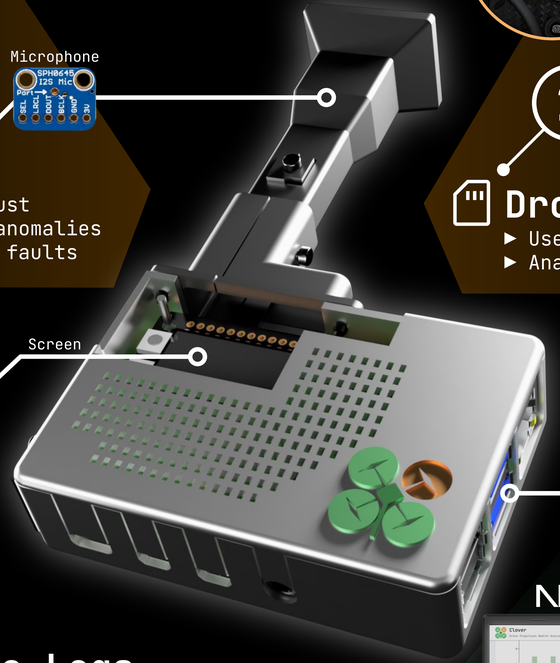
Raspberry Pi 4 Microcontroller

Raspberry Pi is a trademark of Raspberry Pi Ltd



Flight Controller

Microphone



Screen

USB Connection

1 Audio Analysis

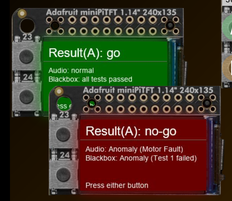
- Records drone audio at low thrust
- Audio transformation isolates anomalies
- Deep neural network identifies faults

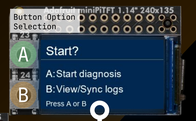
2 Drone Blackbox Data

- Uses accelerometers and gyroscopes
- Analyzes most recent flight data

3 Display Results

- Step-by-step instructions
- Displays diagnostic results





Workflow Map

Sync Logs

- Cloud database stores logs

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4 View Logs

- View logs by pairing to the mobile and web app
- Detailed results for individual components

clover-diagnostics.vercel.app

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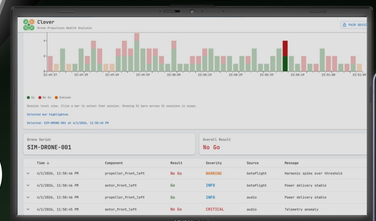
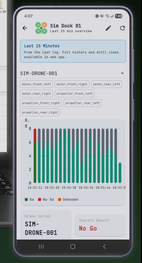
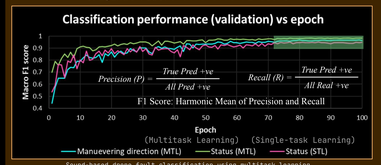


Flutter



Design Verification



Audio Analysis

- *Dataset Drone Audio*

The deep neural network achieved 0.974 F1 score in classifying 9 drone statuses, and 0.994 F1 score in binary classification for Go/No-Go.

- *Our Drone Audio*

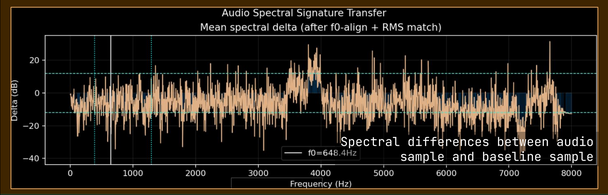
Our drone differs from those used in the dataset, leading to lower accuracy and F1 scores.

~20 baseline samples with ~600 samples total taken from 27 audio recordings of our drone

Metrics for binary Go/No-Go

0.815/1 Accuracy

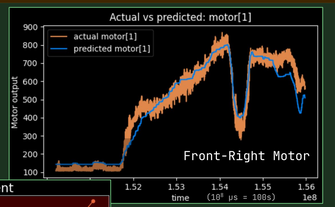
0.715/1 F1 score

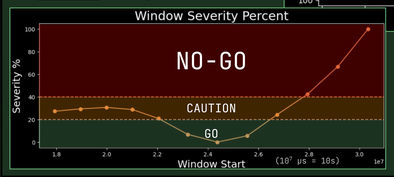


Blackbox Data

We compare the expected motor behaviour with the actual motor behaviour, and compute differences.

~65s of data at a 4kHz recording frequency with ~250µs between each recorded data point, leading to ~260,000 data points.





Metrics for binary Go/No-Go

0.868/1 Accuracy

0.929/1 F1 score

Future Work

- **More Testing:** Test with D. Vision Aerials' drone fleet as a whole; testing in windy / cold conditions
- **Deeper Analysis:** Expand blackbox analytics (e.g., battery monitoring, PID tuning) and improve algorithm speed.
- **Enterprise User System:** User accounts for fleet-wide diagnostic tracking.
- **Additional Sensors:** Extend diagnostics ability using more sensors (e.g., thermal monitoring).